

LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 3

Course Code: CSE276

Course Title: ARTIFICIAL INTELLIGENCE
FOUNDATIONS

Course Outcome: CO3

Paper: Sample Multiple Choice Questions - ANSWER
KEY

Maximum Marks: 40 x 1 = 40

Unit: 3

Unit Title: Intelligent Agents

Unit Mapped RBT: L2 (Understand)

Total Questions: 40

Suggested Duration: 60 Minutes

Answer Grid (Key)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1	A	Q11	D	Q21	A	Q31	C
Q2	B	Q12	B	Q22	C	Q32	A
Q3	C	Q13	D	Q23	B	Q33	A
Q4	D	Q14	C	Q24	D	Q34	D
Q5	B	Q15	B	Q25	C	Q35	B
Q6	D	Q16	A	Q26	D	Q36	C
Q7	A	Q17	B	Q27	A	Q37	C
Q8	C	Q18	A	Q28	B	Q38	B
Q9	C	Q19	D	Q29	D	Q39	D
Q10	A	Q20	C	Q30	B	Q40	A

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Name the element of the PEAS framework that describes the surroundings in which the agent operates.

- A) Environment <-- correct
- B) Performance measure
- C) Actuators
- D) Sensors

Answer: A. The environment is everything the agent acts within and perceives, and its properties decide which agent architecture is adequate.

Q2. Identify the element of the PEAS framework by which an agent acts upon its world.

- A) Sensors
- B) Actuators <-- correct
- C) Environment
- D) Performance measure

Answer: B. Actuators are the means of acting: wheels and grippers on a robot, or a written record and an alert in a software agent.

Q3. State the element of the PEAS framework through which an agent perceives its world.

- A) Actuators
- B) Environment
- C) Sensors <-- correct
- D) Performance measure

Answer: C. Sensors supply the percepts. What they cannot reveal is precisely what makes an environment partially observable.

Q4. Name the component of a learning agent that selects the actions to be taken.

- A) The critic
- B) The learning element
- C) The problem generator
- D) The performance element <-- correct

Answer: D. The performance element is what a non-learning agent consists of entirely; the other three components exist in order to improve it.

Q5. Identify the component of a learning agent that improves the performance element.

- A) The critic
- B) The learning element <-- correct
- C) The problem generator
- D) The actuator set

Answer: B. The critic reports how well the agent is doing, and the learning element uses that report to modify the part that chooses the actions.

Q6. State the term for a rule of the form: if this condition holds, take this action.

- A) A utility function
- B) A percept sequence
- C) A transition model

D) A condition-action rule <-- correct

Answer: D. Condition-action rules are what a simple reflex agent consists of, matching the current percept directly to a response.

Q7. Name the agent that looks up its action in a table indexed by the whole percept sequence.

- A) A table-driven agent <-- correct
- B) A model-based reflex agent
- C) A goal-based agent
- D) A learning agent

Answer: A. A table-driven agent is the direct implementation of the agent function, and it is impractical because the table would be astronomically large.

Q8. Identify the function an agent uses to compare two outcomes that both satisfy its goal.

- A) A goal test
- B) A condition-action rule
- C) A utility function <-- correct
- D) A percept function

Answer: C. A utility function assigns a degree of desirability to each state, which lets a utility-based agent choose between merely satisfactory alternatives.

Q9. State the term for an environment in which the sensors reveal the complete state at every moment.

- A) Partially observable
- B) Deterministic
- C) Fully observable <-- correct
- D) Episodic

Answer: C. Full observability removes the need for the agent to keep any internal record of what it cannot currently see.

Q10. Name the term for an environment whose rules of operation are not known to the agent in advance.

- A) An unknown environment <-- correct
- B) A partially observable environment
- C) A stochastic environment
- D) A continuous environment

Answer: A. An unknown environment is one whose outcomes the agent must discover by acting. This is distinct from partial observability, which concerns perception.

Q11. Identify what the architecture of an agent consists of.

- A) The set of goals the designer has supplied
- B) The table of percept sequences and actions
- C) The rules held in the agent's knowledge base
- D) The computing device together with its sensors and actuators <-- correct

Answer: D. The architecture is the physical platform the agent program runs on. Architecture and program together make up the agent.

Q12. State the term for a multi-agent setting in which all the agents share the same objective.

- A) Competitive
- B) Cooperative <-- correct
- C) Episodic
- D) Deterministic

Answer: B. In a cooperative setting the agents can help one another; in a competitive one an improvement for one is a loss for another.

Q13. Name the multi-agent setting in which one agent's gain is another agent's loss.

- A) Cooperative
- B) Sequential
- C) Static
- D) Competitive <-- correct

Answer: D. Games are the standard example. Competitive settings call for reasoning about what the other agent will choose.

Q14. Identify the agent type that holds an explicit description of the state it is trying to reach.

- A) A simple reflex agent
- B) A table-driven agent
- C) A goal-based agent <-- correct
- D) A model-based reflex agent

Answer: C. Holding the goal explicitly lets the agent consider sequences of actions and choose one because of where the whole sequence leads.

Q15. State the term for an arrangement in which a person reviews or approves the decisions an agent proposes.

- A) Full autonomy
- B) Human in the loop <-- correct
- C) Multi-agent coordination
- D) Reflex control

Answer: B. Keeping a person in the loop is the standard safeguard wherever a wrong decision would be costly and the agent cannot be relied on absolutely.

Q16. Identify the four things on which the rationality of an action depends.

- A) The performance measure, prior knowledge, actions and percepts <-- correct
- B) The sensors, actuators, memory and processor speed
- C) The goal, the utility, the reward and the penalty
- D) The environment, the architecture, the program and the goal

Answer: A. Rationality is judged against what the agent could have known and could have done, not against what turned out to be best in hindsight.

Q17. Explain why rationality is not the same thing as omniscience.

- A) A rational agent always knows every outcome
- B) A rational agent acts best on what it could know <-- correct
- C) A rational agent is never mistaken about anything
- D) A rational agent must perceive the whole state

Answer: B. An omniscient agent would know the actual outcome of every action. Rationality asks only that the expected performance be maximised given the percepts available.

Q18. Distinguish a simple reflex agent from a table-driven agent.

- A) A reflex agent uses rules, a table-driven agent stores every case <-- correct
- B) A table-driven agent uses rules, a reflex agent stores cases
- C) Both of the two store an entry for every percept sequence
- D) Both of the two summarise their behaviour into rules

Answer: A. Condition-action rules compress the behaviour into a manageable set. An explicit table would have to enumerate every possible history, which is impossible.

Q19. Describe what the performance measure of a PEAS specification should be based upon.

- A) The number of actions the agent carries out
- B) The speed at which the agent's program runs
- C) The confidence the agent has in its choices
- D) The state of the environment the agent brings about <-- correct

Answer: D. Measuring the outcome in the world avoids rewarding activity for its own sake, which is how badly written measures produce absurd behaviour.

Q20. Interpret the following PEAS entry: the area of floor left clean for each unit of energy consumed.

- A) It is an actuator of the agent
- B) It is a sensor of the agent
- C) It is a performance measure <-- correct
- D) It is a property of the environment

Answer: C. It states the criterion by which success is judged, which is exactly what the P of PEAS records.

Q21. Predict what a utility-based agent does when no available action fully satisfies its goal.

- A) It selects the action with the highest expected utility <-- correct
- B) It halts and reports that no solution exists
- C) It selects an action entirely at random
- D) It repeats the last action it took before

Answer: A. A utility function grades every outcome rather than merely passing or failing it, so the agent can still make a considered choice among imperfect options.

Q22. Recognize the environment property described: a driving agent whose sensors cannot see round the corner ahead.

- A) Fully observable
- B) Episodic
- C) Partially observable <-- correct
- D) Static

Answer: C. The percepts do not reveal the complete state, so the agent must maintain an internal model of the part of the world it cannot currently see.

Q23. Explain why a learning agent needs to take actions it does not currently believe to be best.

- A) Otherwise its sensors gradually stop functioning
- B) Otherwise it never discovers a better course of action <-- correct
- C) Otherwise its performance measure cannot be computed
- D) Otherwise its actuators become uncalibrated

Answer: B. An agent that only exploits what it already knows never gathers the experience that would reveal something better. The problem generator exists to supply that experience.

Q24. Summarize what the agent architecture and the agent program make up between them.

- A) The environment the agent acts in
- B) The performance measure of the agent
- C) The percept sequence of the agent
- D) The agent itself <-- correct

Answer: D. The program computes an action from a percept; the architecture is the machinery it runs on and through which it senses and acts.

Q25. Distinguish a cooperative multi-agent system from a competitive one.

- A) In a competitive system the agents share an objective
- B) In a cooperative system there is only one agent
- C) In a cooperative system the agents share an objective <-- correct
- D) In a competitive system no agent has any objective

Answer: C. The distinction decides everything about the design: cooperative agents can share information freely, while competitive ones must reason about deception and strategy.

Q26. Explain why communication is needed among the agents of a multi-agent system.

- A) Communication removes the need for any sensors
- B) Communication guarantees the agents will agree

- C) Communication reduces the number of agents needed
- D) No agent can observe the whole state on its own <-- correct

Answer: D. Each agent sees only its own part of the environment. Without exchanging what they observe and intend, their individually sensible choices can conflict.

Q27. Interpret the following as an agent type: a thermostat that switches the heating on whenever the temperature falls below a set point.

- A) A simple reflex agent <-- correct
- B) A goal-based agent
- C) A utility-based agent
- D) A learning agent

Answer: A. The action is a direct function of the current percept, with no memory of what has gone before and no model of consequences.

Q28. Describe what emergent behaviour means in a multi-agent system.

- A) Behaviour copied from one agent by all the others
- B) Group behaviour that no single agent was programmed with <-- correct
- C) Behaviour supplied to every agent by the designer
- D) Behaviour that stops as soon as one agent fails

Answer: B. Simple individual rules can produce complex collective patterns, some of them useful and some, such as congestion or deadlock, distinctly not.

Q29. Explain why a goal-based agent needs a model of how its actions change the world.

- A) It must store every percept it has received
- B) It must compare two outcomes for desirability
- C) It must be able to communicate with other agents
- D) It must judge where a sequence of actions will lead <-- correct

Answer: D. The agent commits to a first action because of the state the whole sequence reaches, and it can only work that out with a model of the effects of acting.

Q30. Predict the effect of giving every agent in a shared space the same fixed priority rule.

- A) The agents always reach the best joint outcome
- B) Symmetrical situations may leave every agent waiting <-- correct
- C) The agents no longer need any sensors at all
- D) The agents will always avoid one another safely

Answer: B. Identical rules applied to a symmetrical situation give identical responses, which is how deadlock arises. Breaking the symmetry is part of coordination design.

Q31. Recognize the collaboration pattern described: the system proposes a shortlist of options and a person makes the final choice.

- A) Full automation with no human involvement
- B) A multi-agent system with competing agents
- C) Human-AI collaboration with a person deciding <-- correct
- D) A simple reflex agent acting on its percepts

Answer: C. The system does the work of narrowing the field while the accountable judgement stays with a person, which is the usual pattern in consequential decisions.

Q32. Explain why a performance measure should be stated in terms of the environment rather than the agent's own behaviour.

- A) Otherwise the agent can satisfy it without doing the task <-- correct
- B) Otherwise the agent cannot perceive its own state
- C) Otherwise the measure cannot be written down
- D) Otherwise the agent needs a utility function

Answer: A. A cleaning agent rewarded for dirt collected can drop dirt and collect it again. Rewarding the cleanliness of the floor cannot be gamed in that way.

Q33. Distinguish the sensors of an agent from its actuators.

- A) Sensors perceive the world, actuators change it <-- correct
- B) Actuators perceive the world, sensors change it
- C) Both of the two perceive the state of the world
- D) Both of the two change the state of the world

Answer: A. The agent function maps what the sensors deliver to what the actuators do, which is why the two are named separately in a PEAS specification.

Q34. Explain what the internal state of a model-based reflex agent is for.

- A) To store every action the agent has ever taken
- B) To hold the utility of every possible outcome
- C) To record the rules supplied by the designer
- D) To track the part of the world the sensors cannot see <-- correct

Answer: D. The state lets the agent behave sensibly when the current percept alone cannot distinguish two genuinely different situations.

Q35. Interpret an environment in which no other agent is pursuing an objective of its own.

- A) It is a cooperative multi-agent environment
- B) It is a single-agent environment <-- correct
- C) It is a competitive multi-agent environment
- D) It is a partially observable environment

Answer: B. Crossword solving is single-agent; chess is multi-agent and competitive. The distinction decides whether the agent must reason about other decision makers.

Q36. Summarize the applications of intelligent agents named on the syllabus.

- A) Image compression and file encryption
- B) Spreadsheet calculation and report printing
- C) Robotic cleaning, warehouse handling and traffic control <-- correct
- D) Database indexing and network routing tables

Answer: C. Any setting in which something must perceive, decide and act over time suits an agent design, and the syllabus names robotics and smart infrastructure in particular.

Q37. Describe what makes a system an autonomous intelligent system.

- A) Its behaviour is fixed entirely by its designer
- B) It operates without any sensors of its own
- C) Its behaviour is shaped by its own experience <-- correct
- D) It requires a person to approve every action

Answer: C. Autonomy is a matter of degree. It is preferred because a designer cannot anticipate every situation the system will meet.

Q38. Explain why full automation is not always to be preferred to human-AI collaboration.

- A) A person is always faster than the agent

B) A person supplies judgement and accountability the agent lacks <-- correct

C) A person can perceive more than any sensor

D) Automation is prohibited in every application

Answer: B. Where an error is costly, or the situation falls outside what the system was built for, a person must be able to intervene and be answerable for the outcome.

Q39. Predict what a learning agent is able to do that an agent with fixed rules cannot.

- A) Perceive parts of the world without sensors
- B) Act without any performance measure at all
- C) Guarantee that its first action is correct
- D) Improve its behaviour from the outcomes it observes <-- correct

Answer: D. A fixed agent repeats the same mistake indefinitely. A learning agent uses the critic's report to change how the performance element chooses.

Q40. Distinguish a goal-based agent from a simple reflex agent by what each considers before acting.

- A) The goal-based agent considers where the action leads <-- correct
- B) The simple reflex agent considers where it leads
- C) Both of the two consider where the action leads
- D) Neither of the two considers the current percept

Answer: A. A reflex agent responds to the situation in front of it. A goal-based agent searches over sequences and picks a first action because of the state it eventually reaches.

----- End of Sample Question Set -----